

Problem Statement No. 8

Document Q&A; RAG Agent for Recipe Generator

The Challenge

Food enthusiasts and culinary professionals often rely on unstructured recipe documents, cookbooks, blogs, and user-submitted notes. Searching across these scattered resources is time-consuming, and it is difficult to adapt recipes based on dietary needs, available ingredients, or cultural preferences. The challenge is not just to retrieve recipes but to intelligently query, adapt, and generate personalized cooking instructions.

The Objective

Build a Document Q&A; RAG (Retrieval-Augmented Generation) Agent that acts as an intelligent Recipe Generator. The solution should:

1. Ingest and index recipe documents from cookbooks, PDFs, blogs, and uploaded notes.
2. Enable conversational Q&A; with natural-language recipe queries.
3. Generate personalized recipes based on available ingredients, dietary restrictions, cooking time, and cuisine preferences.
4. Provide step-by-step cooking guidance, ingredient substitutions, nutritional information, and a shopping list.

Mandatory Tech Stack

IBM Bob / IBM Orchestrate or IBM Langflow, IBM Granite Models, IBM watsonx.ai, IBM Cloud, and Retrieval-Augmented Generation (RAG).